

Mandaloju Pavan Teja

AI Research Engineer | Telangana, India

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PROFESSIONAL SUMMARY

AI Research Engineer with deep expertise in NLP and transformer architectures, fine-tuning LLMs for enterprise applications. Published internal research on model optimisation techniques. Passionate about advancing applied AI within the Indian tech ecosystem.

TECHNICAL SKILLS

Data Engineering: SQL, Airflow, NoSQL, Data Pipelines

NLP & Computer Vision: YOLO, Natural Language Processing, GPT, BERT, Computer Vision

MLOps & Deployment: Azure ML, Model Deployment, MLflow, AWS SageMaker, MLOps

Soft Skills: Adaptability, Project Management, Team Collaboration, Time Management, Attention to Detail

PROJECTS

Real-Time Object Detection for Smart Surveillance

Tech Stack: YOLO, Computer Vision

- Trained a YOLOv8 model on a custom 15K-image dataset for multi-class object detection.
- Achieved mAP@50 of 0.87 with real-time inference at 30 FPS on edge devices.
- Integrated OpenCV pipeline for video stream processing and alert generation.

Medical Image Classification using CNN

Tech Stack: TensorFlow, CNN, Deep Learning

- Built a ResNet-50 transfer learning model for pneumonia detection from chest X-rays.
- Trained on NIH dataset (10K images), achieving 94% accuracy and 0.92 AUC.
- Implemented Grad-CAM heatmaps for model interpretability and clinical validation.

Big Data Customer Segmentation Pipeline

Tech Stack: Python, Spark, Hadoop

- Built a Spark-based ETL pipeline processing 500GB of daily transactional data.
- Applied K-Means and DBSCAN clustering to segment 2M+ customers into 8 cohorts.
- Automated pipeline scheduling using Apache Airflow with email alerting on failure.

EDUCATION

B.Tech – Artificial Intelligence & Machine Learning

JB Institute Of Engineering & Technology | 2025 | CGPA: 8.0

Class XII – State Board of technical education and training

2022 | 57.0%

Class X – SSC

2018 | 85.0%